Research Plan for CSE3000 Research Project

Benchmarking the Unobserved: Applying Informal Benchmarking to the Omitted Variables Bias Model

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Background of the research

In causal inference, the main challenge is that treatment effects are often biased by unobserved confounders – hidden variables that affect both the treatment assignment and the outcome. While Randomised Controlled Trials (RCTs) eliminate this bias, observational studies must rely on sensitivity analysis to bound the potential error. Marginal Sensitivity Models (MSM) bounds this bias by placing a limit on the odds ratio of the propensity score (the probability of being treated), denoted by a sensitivity parameter Γ. To guess a realistic value for this unknown Γ, researchers use "Informal Benchmarking" to see how much the odds of treatment would change based on the observed data.

However, the MSM framework is non-parametric and relies on odds ratios or density ratios (depending on the treatment setting), which can be unintuitive and computationally unstable. An alternative framework is the Omitted Variable Bias (OVB) model, which parametrizes the problem using linear regression. In the OVB model, confounding strength is measured intuitively by partial R^2 values – specifically, the proportion of variance the hidden confounder explains in the treatment (R^2_T) and the outcome (R^2_Y). This project bridges this by doing three things: theoretically mapping the informal benchmarking procedure to the R^2-based OVB framework, implementing the algorithm, and stress-testing its accuracy using synthetic data. Ultimately, this project explores whether this parametric extension generates more intuitive and stable robustness values (RV).

Research Question

The main question of this project is: **Can we apply the idea of informal benchmarking on the omitted variables bias (OVB) model?**

This is reasonable to achieve within the 10-week time frame because it does not require inventing new mathematical frameworks, but rather translating an existing algorithmic procedure (leave-one-out benchmarking) into an existing statistical framework (partial R^2 regression models). I expect to successfully program this translation and show its efficacy using synthetic data.

To structure the research, the main question is broken down into the following sub-questions:

1. **Data Generation:** How can we simulate synthetic datasets with a controlled unobserved confounder, and how do specific properties - such as the true confounding strength (partial R^2) and the correlation between the unobserved and observed covariates - affect the stability of the informal benchmarks? (Success criteria: a synthetic baseline model that correctly hides a variable and allows for parameterized control over confounding strength and covariance).

2. **Algorithmic Translation:** How does dropping observed features iteratively change the partial R^2 bounds of the treatment and outcome in a linear regression setting? (Success criteria: a working Python script that outputs R^2_T and R^2_Y benchmarks for every observed variable).

3. **Evaluation:** How reliably do the calculated informal partial R^2 benchmarks bound the true omitted variable bias? (Success criteria: evaluating the algorithm based on *coverage probability* – the frequency with which the estimated OVB bounds contain the true simulated treatment effect – and analyzing the parameter estimation bias (median vs. true R^2) to determine if the benchmarking procedure leads to accurate robustness conclusions).

Method

To accomplish these aims, the research will be entirely computational and simulation-based.

Tools and Software: I will use Python as the primary programming language. Libraries like `statsmodels`, `numpy` and `pandas` will prove useful for generating the dataset and perform calculations.

Execution Plan: First, I will build a synthetic data generator that creates observed variables (X), an unobserved variable (U), a treatment assignment (T), and an outcome (Y). I will intentionally drop U to simulate unobserved confounding. Next, I will build the benchmarking algorithm that logs the change in R^2_T and R^2_Y. Finally, I will evaluate the algorithm by computing the coverage probability of the estimated sensitivity bounds and analyzing whether the benchmarked Robustness Values (RV) lead to correct conclusions regarding the true simulated bias.

Collaboration: I will work alongside my peer group who are investigating other aspects of sensitivity analysis (e.g., sample sizes, misspecification). We will share literature insights and potentially use the same baseline synthetic data generation scripts to ensure our results are comparable.

Planning of the research project

The timeline below outlines the tasks, meetings, and deadlines for the 10-week quarter. A detailed plan is provided for Week 1 to establish the foundation of the project, followed by the general milestones for the remainder of the quarter.

Week	Planned Activities (Tasks, Meetings, Milestones, Deadlines)
Week 1	**Tasks:** - Read the Informal Benchmarking paper (Baitairian et al., 2025) to understand the high-level goal. - Begin the Chernozhukov (2024) OVB paper. Focus ONLY on the Introduction, the intuition behind the parametric framework, and the definition of the Robustness Value (RV). Do not get bogged down in the complex proofs yet. - Set up Python environment and initialize the GitHub repository. - Draft the Research Plan (this document). **Meetings:** - Group Meeting: Discuss the basic concept of Informal Benchmarking and pool resources for understanding the OVB math.
Week 2	**Tasks:** - Continue the OVB deep dive. Focus specifically on how partial R^2 is calculated and how the sensitivity bounds are mathematically constructed. - Define the exact parameters for the synthetic data generation (varying confounding strength Γ^* and correlation $\rho_{X,U}$) based on the math. - Write the initial Python script to generate the synthetic datasets (X, U, T, Y). **Meetings:** - Supervisor meeting: Finalize data generation parameters and ask specific questions to clarify the OVB bounds.

Week	Planned Activities (Continued)
Week 3	**Tasks:** - Finish digesting the necessary components of the OVB paper. Map the mathematical bounds to actionable algorithmic steps. - Run baseline linear regression on the synthetic data, explicitly dropping U to observe the simulated omitted variable bias. - Draft the Introduction and Formal Problem Description sections of the paper. **Meetings:** - Peer group meeting: Troubleshoot initial data generation and align on the algorithmic translation of the OVB math.
Week 4	**Tasks:** - Code the leave-one-out informal benchmarking algorithm explicitly adapted for the OVB (R^2) framework. - Execute the algorithm across all simulated datasets to extract and log the R_T^2 and R_Y^2 shifts. - Prepare slides and preliminary data visualizations for the midterm presentation. **Meetings:** - Pre-midterm meeting to discuss the raw benchmark outputs and verify the algorithm is working.
Week 5	**Tasks:** - Write the evaluation script to compute the *coverage probability* of your estimated bounds. - Calculate the bias between your estimated Robustness Values (RV) and the true simulated bias. - Draft the Experimental Setup section. **Deadlines:** - Midterm Presentation. - Midterm Poster. **Meetings:** - Midterm evaluation meeting to readjust scope based on initial evaluation metrics.
Week 6	**Tasks:** - Generate final OVB sensitivity contour plots showing the robustness and non-robustness areas based on your results. - Draft the Results section, integrating the coverage probability and RV accuracy findings. - Combine existing sections into a cohesive Draft v1. **Deadlines:** - Submit Paper Draft v1. **Meetings:** - Supervisor meeting to receive feedback after midterms.
Week 7	**Tasks:** - Incorporate Draft v1 feedback from peers and supervisor. - Write the Discussion section, specifically detailing where the benchmarking algorithm failed or succeeded based on the data properties (e.g., when X and U were highly correlated). - Write the Conclusions, Future Work, and Responsible Research sections.

Week	Planned Activities (Continued)
Week 8	**Tasks:** - Finalize full manuscript, verify all LaTeX equations, and format the bibliography. - Begin drafting the final poster summarizing the research metrics and contour plots. **Deadlines:** - Submit Paper Draft v2. **Meetings:** - Responsible professor meeting to discuss Draft v2 and overarching conclusions.
Week 9	**Tasks:** - Implement final revisions to the paper based on professor's feedback. - Prepare slides for final presentation. - Clean up Python code, document functions, and prep GitHub repository for public release. **Meetings:** - Final supervisor meeting with a dry-run presentation.
Week 10	**Tasks:** - Polish and export the final version of the poster. - Rehearse talking points for the Q&A. **Deadlines:** - Submit the final version of the Research Paper. - Final presentation. - Poster Q&A session with the examiner.